

Problem 1:

a. $[r, b, b, r, \text{grn}, \text{gray}]$

	r	b	grn	g
OHE $\rightarrow$	1	0	0	0
	0	1	0	0
	0	1	0	0
	1	0	0	0
	0	0	1	0
	0	0	0	1

b. $\hat{r} = \frac{2}{6} = \frac{1}{3}$

$\hat{b} = \frac{2}{6} = \frac{1}{3}$

$\hat{\text{green}} = \frac{1}{6}$

$\hat{\text{gray}} = \frac{1}{6}$

c. $X = [1, 3, 3, 6, 11]$

$F(x) = P(X \leq x)$

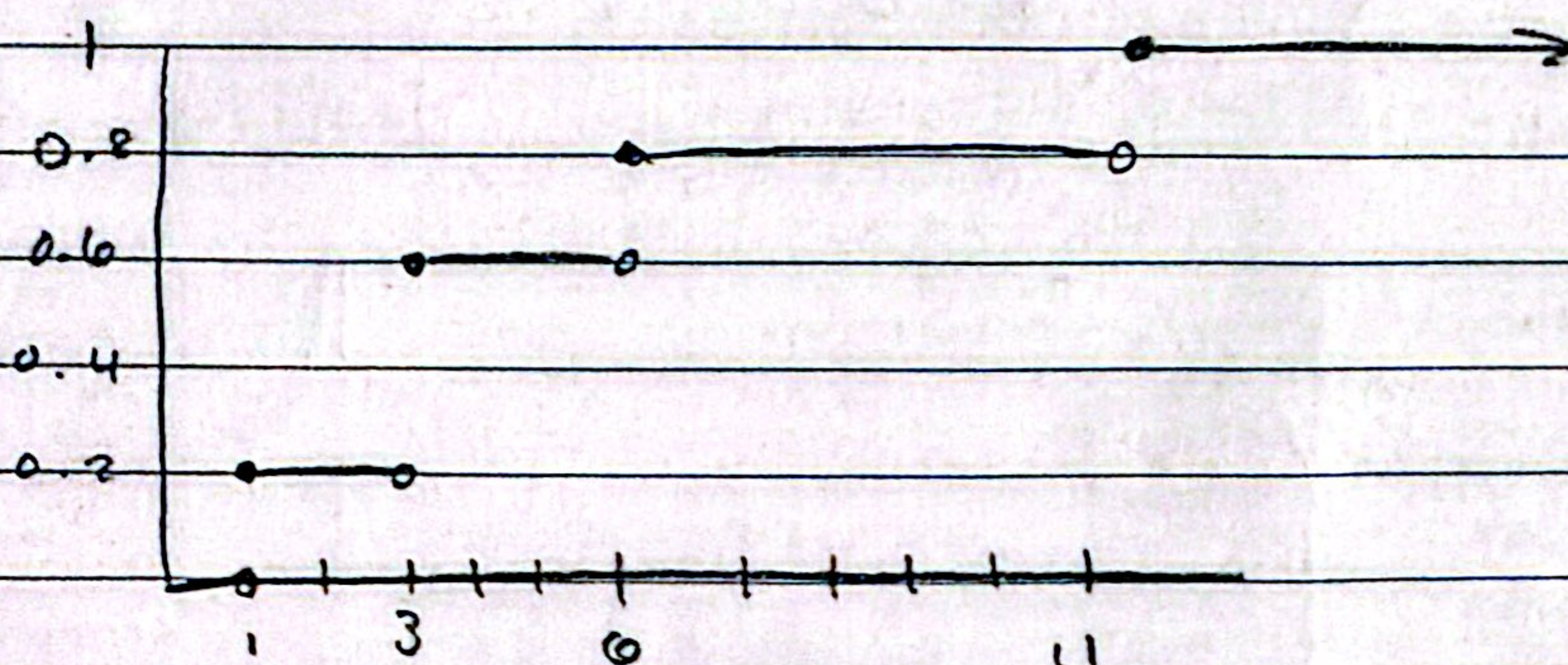
$P(X \leq 1) = \frac{1}{5}$

$P(X \leq 3) = \frac{3}{5}$

$P(X \leq 6) = \frac{4}{5}$

$P(X \leq 11) = 1$

$$F(x) = \begin{cases} 0, & \text{if } x < 1 \\ 0.2 & \text{if } 1 \leq x < 3 \\ 0.6 & \text{if } 3 \leq x < 6 \\ 0.8 & \text{if } 6 \leq x < 11 \\ 1 & \text{if } x \geq 11 \end{cases}$$



Problem 2:

a) $Y = a + bX$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(a + bX \leq y) \\ &= P(X \leq \frac{1}{b}(y-a)) \\ &= F_X\left(\frac{1}{b}(y-a)\right) \leftarrow F_Y(y) \end{aligned}$$

$$\begin{aligned} \frac{d}{dy} F_Y(y) &= \frac{d}{dy} F_X\left(\frac{1}{b}(y-a)\right) \\ &= \frac{1}{b} f_X\left(\frac{1}{b}(y-a)\right) \leftarrow f_Y(y) \end{aligned}$$

$$X = [1, 3, -1]$$

b) $\bar{X} = 8/3$

c) $1 + 9 + 16 = \bar{X}^2 = 26/3$

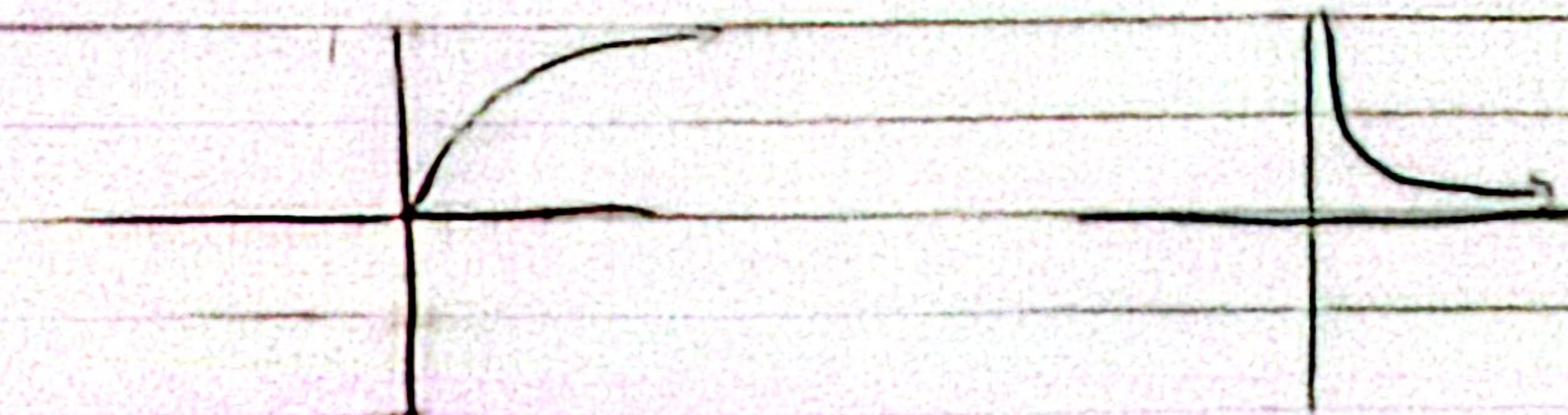
d) $(\bar{X})^2 = 64/9 \approx 7.1$

$$\bar{X}^2 = 26/3 \approx 8.6$$

no, they don't equal each other

a)
$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

density $\frac{d}{dx} 1 - e^{-x} = e^{-x}$



Problem 3:

b)

$$\begin{aligned} P(X \geq 3) &= 1 - P(X \leq 3) \\ &= 1 - F(3) \\ &= 1 - (1 - e^{-3}) \\ &\approx 0.0498 \end{aligned}$$

$$P(X \leq 35) = 1 - e^{-35} \approx 1$$

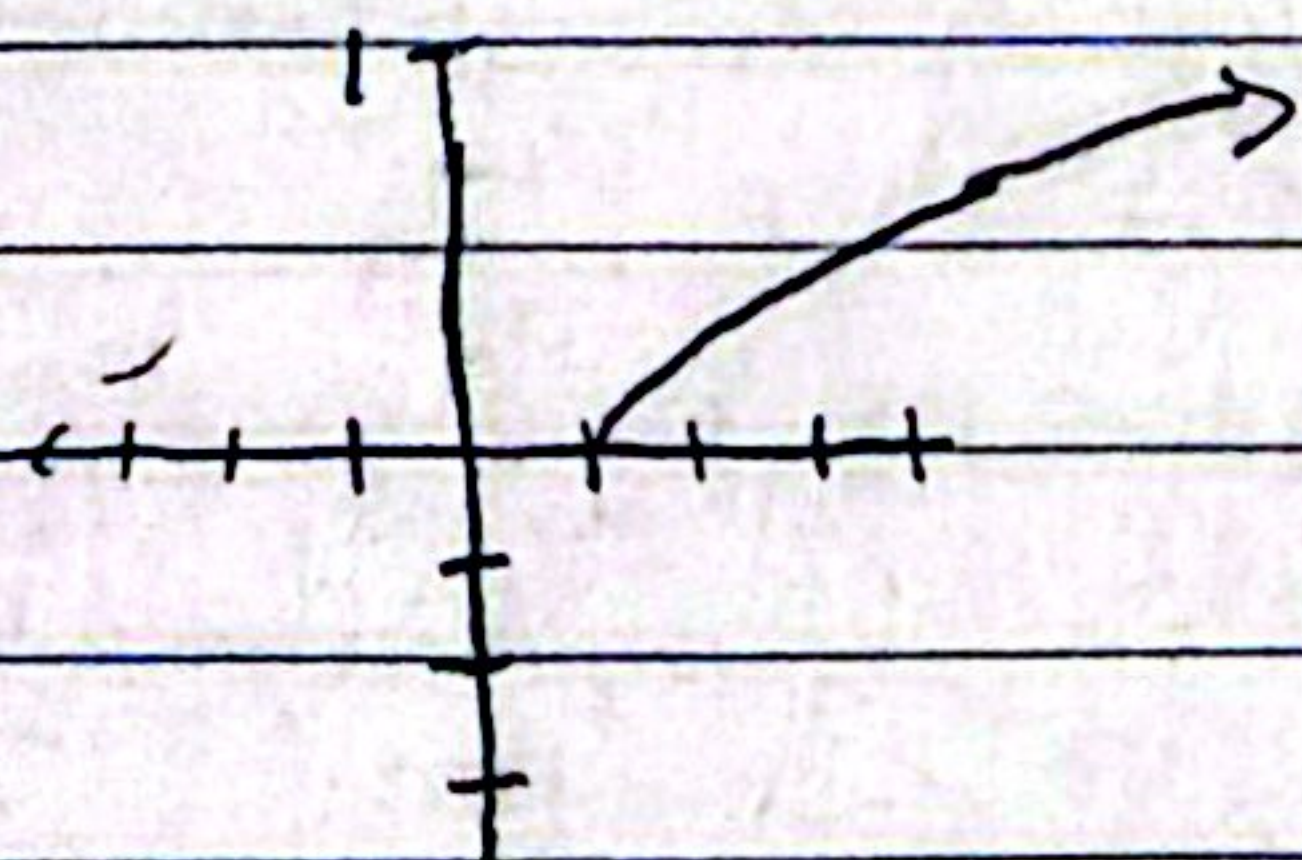
c) $F_Y(y) = P(Y \leq y)$

$$= P(1 + 3X \leq y)$$

$$= P(X \leq \frac{1}{3}(y-1)) = F_X(\frac{1}{3}(y-1)) = 0 \text{ at } y=1$$

$$F_Y(y) = \begin{cases} 0 & y < 1 \\ 1 - e^{-\frac{1}{3}(y-1)} & y \geq 1 \end{cases}$$

↑
Plug
↙

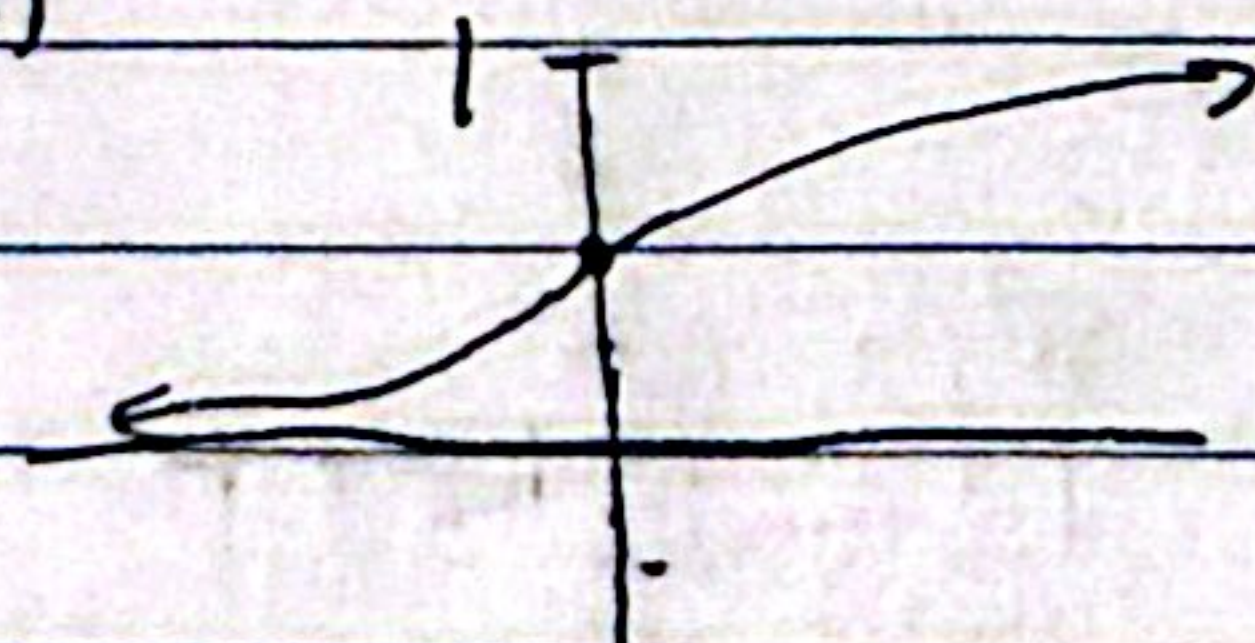


$$\begin{aligned} P(Y \leq 10) &= 1 - e^{-\frac{1}{3}(10-1)} \\ &= 1 - e^{-3} \\ &\approx 0.95 \end{aligned}$$

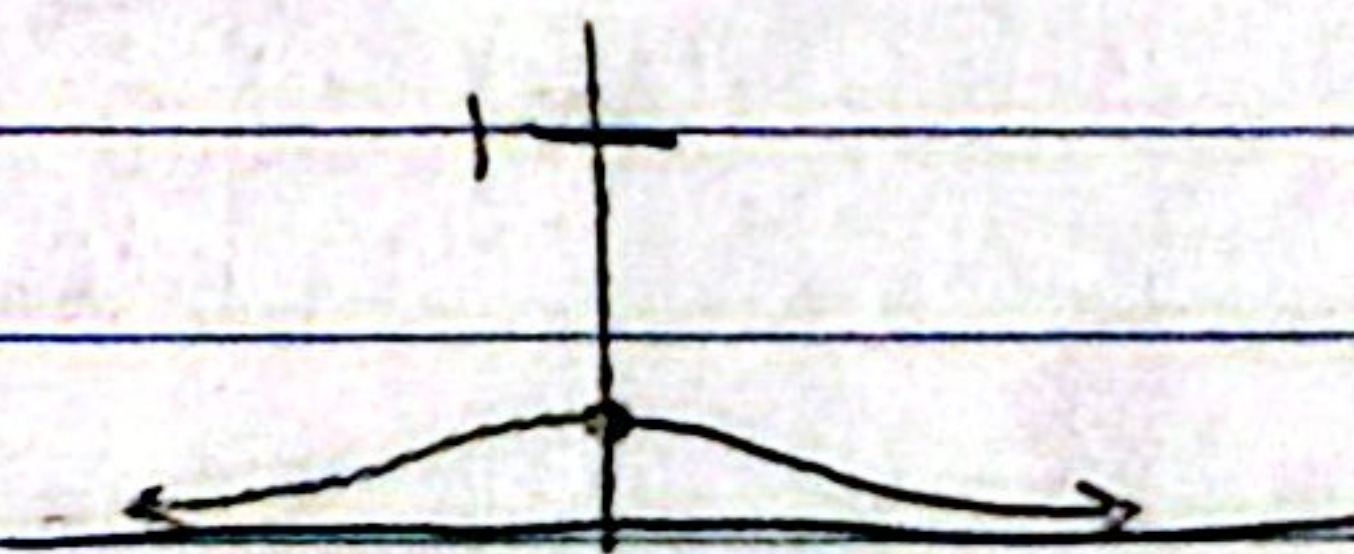
Problem 4:

$$\frac{1}{1+e^{-x}}$$

a) $F(x)$



$$f(x) = \frac{d}{dx} (1+e^{-x})^{-1} = \frac{e^{-x}}{(1+e^{-x})^2}$$

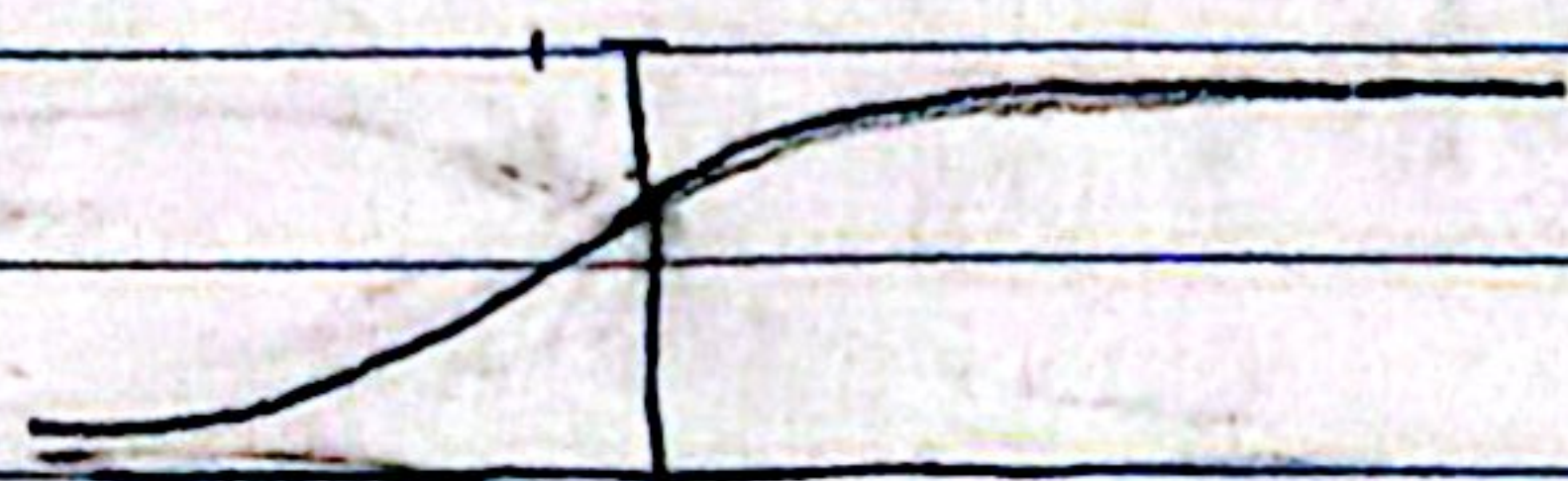


b) $P(X \geq 0.8) = 1 - \frac{1}{1+e^{-0.8}} \approx 0.310$

$$P(X \leq 0.3) = \frac{1}{1+e^{-0.3}} \approx 0.574$$

$$\begin{aligned}
 c) \quad F_Y(y) &= P(Y \leq y) \\
 &= P(2X - 1 \leq y) \\
 &= P(X \leq \frac{1}{2}(y+1)) = F_X
 \end{aligned}$$

$$F_Y(y) = \frac{1}{1 + e^{\frac{1}{2}(y+1)}}$$



$$\begin{aligned}
 P(Y < 0) &= \frac{1}{1 + e^{\frac{1}{2}(0+1)}} \\
 &\approx 0.622
 \end{aligned}$$

Problems 5:

$$a) \quad F_X(x) = 1 - e^{-x}$$

$$\begin{aligned}
 P(X \geq x) &= \frac{1}{2} = 1 - e^{-x} \\
 \ln\left(\frac{1}{2}\right) &= -e^{-x} \\
 x &= \ln(2) = \text{median}
 \end{aligned}$$

$$b) \quad F'_X(x) = \frac{1}{1 + e^{-x}}$$

$$P(X \geq x) = \frac{1}{2} = \frac{1}{1 + e^{-x}}$$

$$1 + e^{-x} = 2$$

$$e^{-x} = 1 \quad x = 0 \leftarrow \text{median}$$

$$c) \quad U = F(x)$$

$$U = 1 - e^{-x}$$

$$U - 1 = -e^{-x}$$

$$1 - U = e^{-x}$$

$$\ln(1 - U) = -x$$

$$x = -\ln(1 - U) = F^{-1}(u)$$

$$d) \quad U = F(x)$$

$$U = \frac{1}{1 + e^{-x}}$$

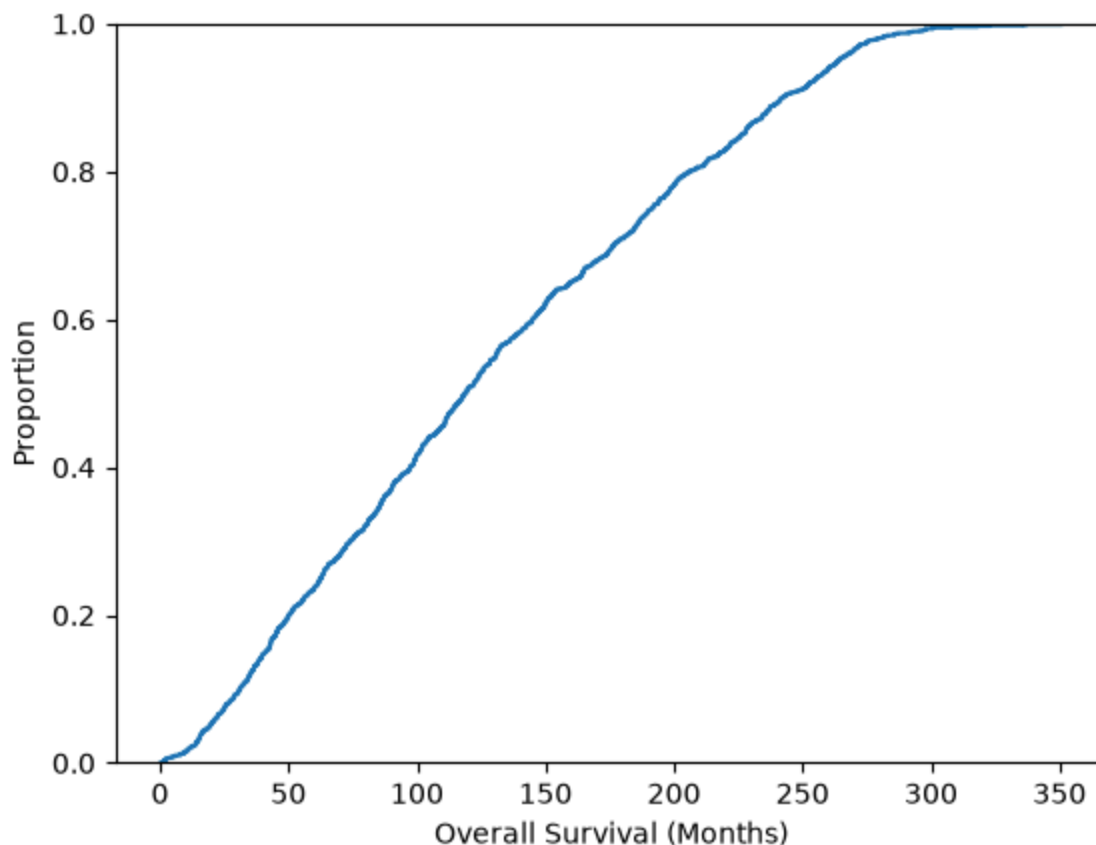
$$\begin{aligned}
 (1 + e^{-x}) &= \frac{1}{U} \\
 e^{-x} &= \frac{1}{U} - 1
 \end{aligned}$$

$$\ln\left(\frac{1}{U} - 1\right) = -x$$

$$x = -\ln\left(\frac{1}{U} - 1\right) = F^{-1}(u)$$

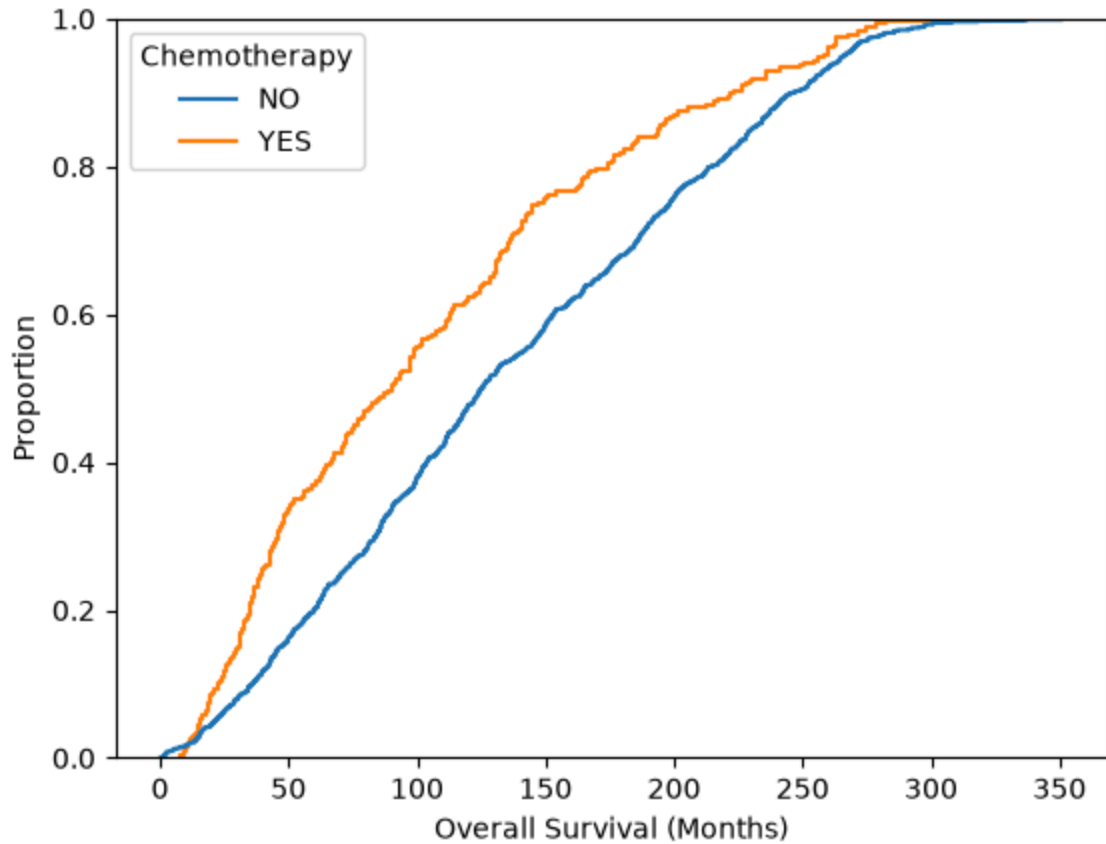
```
In [1]: import pandas as pd
import numpy as np
import scipy.stats as stats
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [5]: # a
metabric = pd.read_csv("data/metabric.csv")
sns.ecdfplot(data = metabric, x = "Overall Survival (Months)")
plt.show()
```



```
In [ ]: # b
sns.ecdfplot(data = metabric, x = "Overall Survival (Months)",
             hue = "Chemotherapy")
plt.show()
print(metabric.groupby("Chemotherapy")
      ["Overall Survival (Months)"].describe()[
      ["count", "mean", "25%", "50%", "75%"]])

# based on the graph and the summary statistics, patients
# who received chemotherapy had a lower median and when
# analyzing the graph, the patients who received chemotherapy
# were to the left of the patients who did not receive chemotherapy,
# meaning that their survival months was lower. Therefore,
# chemo patients survive for a shorter time.
```



	count	mean	25%	50%	75%
Chemotherapy					
NO	1057.0	135.737496	70.533333	125.266667	198.300000
YES	286.0	104.542308	39.441667	90.266667	146.366667

```
In [ ]: # c
cols = ["Age at Diagnosis", "Lymph nodes examined positive",
        "Nottingham prognostic index", "Tumor Size", "Tumor Stage"]
print(metabric.groupby("Chemotherapy")[cols].mean().round(2).T)
print(metabric.groupby(["Tumor Stage", "Chemotherapy"])
      ["Overall Survival (Months)"].agg(["count", "median"]).round(1))
```

I think that it is difficult to determine whether chemotherapy is effective treatment based on the data. Some of the reasons that make it difficult to determine the effectiveness of chemotherapy are the fact that the patients who received chemotherapy were already at stage 2 or worse. The mean of the size and stage were much higher than the patients who did not receive chemotherapy. One example is that when we look at stage 2, the difference between median survival months is only about 12 because we have a larger sample size and the cancer itself is not as serious as stage 3 or 4. Overall, we were looking at Chemotherapy as predictor of survival months, but it is not the only factor that affects survival months. In this case we are not adjusting for the other factors that affect survival months, so we cannot determine if its an effective treatment or not.

Chemotherapy	NO	YES
Age at Diagnosis	63.40	50.22
Lymph nodes examined positive	1.26	4.09
Nottingham prognostic index	3.78	5.00
Tumor Size	24.08	32.08
Tumor Stage	1.64	2.15

		count	median
Tumor Stage	Chemotherapy		
0.0	NO	2	113.9
1.0	NO	441	159.0
	YES	17	76.7
2.0	NO	555	111.0
	YES	212	98.6
3.0	NO	55	77.2
	YES	54	46.7
4.0	NO	4	107.8
	YES	3	38.1